Greenwich - Grown-ups

UK Family Tour

The Island That Set the World's Clock

Before you climb the hill, a thought to carry up it. You are about to spend a day in the place where a small, damp island off the edge of Europe turned itself into the most powerful nation on Earth, and where, as a direct result, the whole planet still agrees on what time it is and where it is.

Those two facts are the same story. Britain had no great forests of gold, no vast fertile plains, no obvious reason to matter. What it had was the sea, on every side. So it made ships its walls, its roads, and its fortune. For roughly three centuries, British vessels carried tea, spices, cotton and sugar around the globe, and the Royal Navy made sure they arrived. By the nineteen twenties, the empire this built ruled about a quarter of the world's land and a quarter of its people. That is a staggering achievement and a troubling one at once, resting as it partly did on conquest and on the trade in enslaved human beings, and we will not pretend otherwise today.

But sailing the whole world safely meant solving a problem that had baffled everyone for centuries. How do you know where you are when there is nothing but water in every direction? The answer, worked out on this very hill, tied the measurement of place to the measurement of time so tightly that they became a single thing. Which is why the map of the world and the clock of the world both begin here, in Greenwich, at zero. Let's go and stand on it.

Wooden Walls and the Wealth of the World

Look at a map of Britain and you notice the thing its people noticed centuries ago. Nowhere on the island is more than about seventy miles from the sea. For an ambitious nation that is either a prison or a launch pad, and the British chose launch pad.

The turning point most people fix on is fifteen eighty eight, when a small, nimble English fleet, helped by ferocious weather, saw off the enormous Spanish Armada. A Tudor writer called the navy the nation's wooden walls, and the idea stuck. An island that cannot easily be invaded, and that can send armed ships anywhere in the world, holds a peculiar kind of power. It can reach out without being reached.

What followed was less heroic and more commercial. Trading companies, above all the East India Company, grew into something between a business and a government, with their own armies and their own territories. Ships left British ports with manufactured goods and came home with tea from China, cotton and spices from India, sugar from the Caribbean. That sugar is the hard part of the story. For well over a century British ships also carried enslaved Africans across the Atlantic in appalling numbers, and much of the wealth on display in grand British buildings was grown from that cruelty, before Britain finally abolished the slave trade in eighteen oh seven and slavery itself across most of the empire in eighteen thirty three.

There is a reason so much of this ran through Greenwich in particular. This bend in the river Thames, just downstream of London, was where the naval story lived. Kings and queens kept a palace here. The dockyards that built and repaired warships lay close by along the river. And the observatory on the hill existed to serve the fleet, which is why the machinery of empire, the ships, the science, the grand architecture, all ended up clustered on this one green slope where you are standing.

Hold both halves in your head, because Greenwich holds both too. This was a place of dazzling skill, daring, and invention. It was also the command centre of an empire whose reach was as heavy for the conquered as it was glorious for the conquerors. The great sailing ship at the bottom of the hill and the observatory at the top are two faces of one ambition. To go everywhere, know everything, and bring it home.

Lost at Sea, and the Clock That Found the Way

Here is the puzzle that ruled everything. Out of sight of land, on an empty ocean, how do you know where you are?

Half the answer was easy and ancient. Sailors could measure how high the sun climbed at noon, or find the Pole Star at night, and from that work out their latitude, meaning how far north or south they were. The Greeks could do it. The trouble was the other direction. Longitude, how far east or west you had travelled, left no mark in the sky that anyone could read. Ships crossed oceans essentially guessing, and when they guessed wrong they ran out of water, or struck rocks in the dark, sometimes taking hundreds of lives at once.

The clever insight was that east and west are really a question about time. The Earth turns once in twenty four hours, so every hour the sun sweeps through a slice of the globe. If you could somehow know the exact time back at your home port, and compare it with the local time where your ship actually floated, measured from the sun overhead, the gap between the two would tell you precisely how far around the world you had come. One hour of difference means fifteen degrees of longitude. The whole problem shrank to a single maddening requirement. A clock that could keep perfect time on a heaving, salty, freezing ship, when the finest clocks of the age lived indoors on solid ground and still drifted.

Solving that, here at Greenwich, is one of the great stories of stubbornness in the history of science, and you will hear it in full up the hill. For now, just hold the beautiful strangeness of it. To answer the question where am I in space, humanity first had to master the question what time is it. Place and time became the same measurement. And because the answer was perfected here, this hill became the point that both of them are counted from. When your phone back in Texas shows the time, it is quietly counting hours from this courtyard, whether it knows it or not.

What to Watch For on the Hill

So here is the shape of your day, and a few things worth watching for as you walk it.

Start low, by the river, with the ship. She is the last of the great tea clippers, the racing machines of the sailing age, and she marks the very end of the story, the moment steam and steel finally beat the wind. Notice that they have raised her right off the ground so you can walk beneath her hull. Almost nobody in history got to see a working ship from below.

Then climb. The grand pale buildings you pass through were a royal palace, and later a home for old and injured sailors, a nation saying thank you in architecture to the men who crewed its ships. Higher up sits the observatory, small and businesslike, built not to gaze dreamily at the stars but to chart them precisely enough to save lives at sea.

At the top, three things to find. A brass line set in the ground, which is zero longitude, the seam of the world's map, with one hemisphere under each foot. A bright red ball on a mast, which since the eighteen thirties has dropped at one o'clock every afternoon so that captains on the river below could set their clocks. And a set of extraordinary machines in glass cases, the timekeepers that solved the whole puzzle.

Watch, too, for the moment the modern satellite line and the old brass line don't quite agree, a lovely reminder that even zero has a history.

Before you leave the summit, turn around and take in the view, because it tells the whole story in one glance. Below you sit the perfect white classical buildings of the old naval college. Beyond them curls the river that carried the ships. And rising on the far bank stand the glass towers of Canary Wharf, one of the great money centres of the modern world, built on the very docks where those empire cargoes once landed. Three ages of British power stacked in a single frame, from sail, to steam, to spreadsheets.

One island, one hill, and the whole world's map and clock. Off you go.

Cutty Sark: The End of the Age of Sail

Cutty Sark, launched at Dumbarton in 1869, was built at precisely the wrong moment, which is what makes her poignant. She was designed as the ultimate tea clipper — an extreme sailing machine meant to win the annual race home from China, where the first cargoes of the season commanded premium prices and the races made front pages. But the Suez Canal opened the very year she launched, giving steamships a shortcut sail couldn't use, and the China tea trade abandoned sail within a decade. Her greatest years came instead on the Australian wool run, where in the 1880s, under the hard-driving Captain Woodget, she posted passages no rival could match — Sydney to London in as little as seventy-three days — with recorded runs over three hundred sixty nautical miles in a day. The name comes from Robert Burns's poem Tam o' Shanter: the witch Nannie wears a "cutty sark," a short shift, and the figurehead shows her clutching a horse's tail, snatched from Tam's mare as he fled. The 2007 fire, which broke out during conservation, spared most of the original fabric only because roughly half of it had been removed offsite. The current display, with the hull raised so you walk beneath the copper-clad keel, is controversial among purists — but standing under three thousand square meters of Victorian naval architecture makes the argument feel academic.

> Related listening: Turner painted sail giving way to steam; Cutty Sark lived that same ending. Coming up — "Turner's Temeraire and Van Gogh's Absent Sunflowers" (National Gallery, Day 5).

The Rudder Epic and the Clipper Economy

To understand why anyone built a ship this extreme, follow the money. In the tea trade, the first cargo of the new season's crop to reach London fetched a premium, so speed was cash, and the annual dash home became a public spectacle, wagered on and reported like a Derby. To win it, builders pushed a radical design. Cutty Sark was a composite ship, an iron skeleton planked with wood and sheathed below the waterline in copper, which stayed smooth and fast where a wooden hull would foul with weed. Light, strong, and driven by acres of sail, she was a thoroughbred bred for one race.

That race defines her, and in 1872 it produced one of the great sea stories. She left China alongside her fierce rival Thermopylae, and by the Indian Ocean she was some four hundred miles ahead. Then, in a gale after the Sunda Strait, her rudder tore away and sank. The owner's brother, aboard as passenger, urged Captain George Moodie to put in at Cape Town for repairs. Moodie refused. He would not lose that lead to the harbour.

So they built a rudder at sea. The carpenter, Henderson, fashioned one from spare spars and iron, and here fortune turned strange, because among two stowaways discovered after leaving China, one was a blacksmith. He rigged a forge on the pitching deck to hammer out the ironwork, and in the chaos the brazier of hot coals overturned and burned Moodie's own son. Six days of this, in heavy seas, and they shipped their improvised rudder and sailed on. Cutty Sark reached London only about a week behind Thermopylae, having crossed half the world jury-rigged. Henderson got a fifty pound bonus. The feat, not the loss, became the legend, which tells you something about the culture of sail. Speed mattered, but so did the refusal to quit. Steam would soon make all of it obsolete, but nothing steam ever did could match a story like that.

Afterlives: Ferreira, Whisky and the Figureheads

Great ships rarely end grandly, and Cutty Sark's survival is almost accidental. By 1895, unable to compete with steam, she was sold to a Portuguese firm, Ferreira and Company, and renamed simply Ferreira. Her new crews called her affectionately the Pequena Camisola, the little shirt, a neat translation of that Scots name. For nearly thirty years she carried general cargo around the Atlantic, dismasted once, re-rigged, unglamorous but still working, the very last clipper of her kind afloat.

Her rescue came by chance. In 1922, battered by a Channel storm, she put into Falmouth, where a retired sea captain named Wilfred Dowman recognised her at once. He bought her, for three thousand seven hundred and fifty pounds, funded largely by his wife Catharine, a Courtauld heiress, and restored her as a training ship. In 1954 she was moved into the purpose-built dry dock here at Greenwich, where she sits today, the only surviving tea clipper in the world.

Her fame had already outgrown her hull. In 1923, the London wine merchants Berry Brothers and Rudd wanted a Scotch blend to smuggle into a Prohibition-era America, and they named it after the ship, then freshly in the news back in home waters. The Cutty Sark whisky, with its yellow label and little clipper, made the name a household word on two continents, which is why many people know the drink long before the vessel.

Before you leave, go down into the gallery beneath the raised hull and meet a peculiar crowd. Ranged along the walls is the Long John Silver Collection, more than a hundred ships' figureheads, the largest such collection anywhere. They were gathered by an eccentric businessman, Sydney Cumbers, who lost an eye as a boy, wore a piratical eye patch, and styled himself Long John Silver. He gave them to the ship in 1953. Roman soldiers, admirals, maidens and reformers, all carved to ride the bows of vessels now long gone, they gaze out from under the last clipper like the ghosts of a whole vanished fleet.

The Prime Meridian: How Greenwich Became Zero

The brass-and-steel line in the Observatory courtyard marks a diplomatic settlement as much as a scientific fact. Any meridian can be zero — the choice is pure convention — and through the nineteenth century, national maps used national zeros: Paris, Cadiz, Washington. What forced a decision was the railway and the telegraph, which made local solar time untenable. The International Meridian Conference met in Washington in October 1884, and Greenwich won for a blunt commercial reason: roughly two-thirds of the world's shipping tonnage already navigated with charts based on Greenwich, largely thanks to the Royal Observatory's Nautical Almanac, published since 1767. The vote was twenty-two to one — San Domingo opposed, France and Brazil abstained, and France clung to the Paris meridian for maps into the twentieth century, defining it in the interim as "Greenwich minus nine minutes twenty-one seconds," which is a very French form of surrender. One nice modern wrinkle: your phone's GPS will show you crossing zero longitude about a hundred meters east of the courtyard line. That's not an error in the line — satellite-era geodesy uses a smoothed model of the Earth rather than local astronomy, so the "real" zero drifted. The Observatory itself, founded by Charles the Second in 1675, was created for exactly this enterprise: fixing the stars precisely enough that sailors could find themselves. Which brings you to Harrison.

Railway Time and the Invention of Now

Before the railways, there was no such thing as the time. Every town kept its own noon, set by the sun crossing its own sky, so Bristol ran about ten minutes behind London, and nobody minded, because nobody could travel fast enough for it to matter. Then came the train. Suddenly a timetable that used local time in every station was a recipe for missed connections and collisions, and in the 1840s Britain's railways simply imposed London time along the whole network. They called it railway time, and the telegraph, which could send a time signal down a wire in an instant, made it enforceable. Greenwich time spread out along the rails until the whole country ticked together.

That is the quiet revolution this courtyard represents. Standard time is younger than the steam engine, and it was born of engineering necessity, not philosophy. The international version arrived in 1884, when the world agreed to count its time zones from this hill, each a neat hour ahead of or behind the one beside it. The green laser the Observatory shines north at night traces the very line that organises the planet's clocks.

You'll hear two names for it, and the difference is worth knowing. Greenwich Mean Time was defined by the stars, by astronomers here timing the moment particular stars crossed the meridian. Modern global time is called Coordinated Universal Time, or UTC, and it's kept not by the heavens but by atomic clocks, hundreds of them, averaged around the world, accurate to billionths of a second. The snag is that the Earth's spin is very slightly irregular and gradually slowing, so atomic time and astronomical time drift apart. To keep them aligned, timekeepers occasionally insert a leap second, an extra tick added to a minute, so that once in a while a day officially contains a second more than usual. It is a small, strange act of bookkeeping, patching the perfect clock of physics onto the wobbling clock of the turning Earth, and it all traces back to zero, right here.

The French Sulk and the Two Zeros

Not everyone bowed to Greenwich gracefully. At the 1884 conference, France abstained rather than vote for a British zero, and the resentment ran deeper than pride. There had been an implicit horse-trade in the air, that the world might adopt Greenwich for longitude if the English-speaking powers embraced the elegant French metric system for measurement, and when Britain and America declined to go fully metric, French delegates saw no reason to hand London the prime meridian for nothing. So France kept measuring longitude from Paris for decades. Astonishingly, French law defined its national time until 1911 not as Greenwich time, which would have been an admission, but as Paris Mean Time retarded by nine minutes and twenty-one seconds, which is exactly Greenwich time wearing a disguise. It was a very Gallic way to lose an argument without ever saying the word Greenwich.

Then there is the puzzle that delights every visitor with a smartphone. Stand on the historic brass line, open a satellite map, and your device will place true zero longitude about a hundred metres to the east, out in the park. Neither is wrong. The original meridian was fixed by astronomy, by a telescope aimed straight up along the local vertical, which a plumb line defines using gravity. But the Earth is lumpy, and the pull of unevenly distributed rock beneath your feet tugs that plumb line slightly off from the planet's true centre, an effect called the deflection of the vertical. Satellite systems ignore local gravity entirely and reference a smooth mathematical Earth, the model called WGS 84 that underlies GPS. The gap between the astronomer's zero and the satellite's zero is that hundred-odd metres.

Which points to the real Greenwich legacy. This place never had gold mines or great harbours. What it exported to the world was standards, the shape of a second, the grid of latitude and longitude, the fiction of a shared now. Empires trade in goods, but the most enduring thing shipped from this hill was pure agreement, an invisible line that everyone, even the French, eventually agreed to stand on.

Harrison and the Longitude Prize

The four timekeepers displayed in the Observatory — H1 through H4 — are among the most consequential machines ever built, and the story behind them is a genuine epic of institutional resistance. After the Scilly naval disaster of 1707 drowned around two thousand men, Parliament's Longitude Act of 1714 offered up to twenty thousand pounds — several million in today's money — for a practical method of finding longitude at sea. The astronomical establishment assumed the answer would come from the stars, via the lunar distance method. John Harrison, a self-educated Yorkshire carpenter and clockmaker, bet instead on the "impossible" solution: a clock accurate enough to carry home-port time across an ocean, despite rolling decks, salt air, and temperature swings that wrecked every pendulum and lubricant. His answers were mechanical poetry — H1's linked balances that cancel a ship's motion, H3's bimetallic strip and caged roller bearings, both his inventions and both still in use in modern engineering — culminating in H4 of 1759, a five-inch watch that lost seconds, not minutes, crossing the Atlantic. The Board of Longitude, dominated by astronomers with a rival method, paid out grudgingly and piecemeal; Harrison was in his eighties before King George the Third personally intervened — reportedly declaring "By God, Harrison, I will see you righted" — and Parliament paid. The three big clocks still run. H4 is kept stopped: running it would wear away the very mechanism that changed navigation forever.

The Rival Method and a Conflicted Judge

Harrison's clocks are only half the story, because he was fighting a rival method backed by the entire scientific establishment. It was called lunar distance, and it was genuinely ingenious. The Moon moves against the background stars a little faster than the stars themselves, roughly its own width every hour, so the sky becomes a vast, slow clock. Measure the angle between the Moon and a chosen star very precisely, then consult tables predicting where the Moon should be at each moment of Greenwich time, and you could deduce the home time, and from that your longitude. It needed no expensive machine, only a sextant, a steady hand, and a formidable appetite for arithmetic, about four hours of calculation per fix in the early days.

The champion of this method was Nevil Maskelyne, who became Astronomer Royal here at Greenwich and, crucially, sat on the Board of Longitude that judged Harrison's claim. Maskelyne compiled the Nautical Almanac, first published in 1767, the very tables that made lunar distance practical, so the man assessing whether a clock deserved the prize had built his own reputation on the competing technique. When Harrison's watch was sent for sea trial to Barbados in 1764, Maskelyne sailed out to conduct astronomical observations there too, and Harrison's supporters cried foul, arguing the referee was also playing for the other team. The watch performed superbly, losing only a handful of seconds across the Atlantic, comfortably inside the prize's demanding limit.

Yet the Board would not simply pay. They demanded Harrison hand over his masterpiece, H4, and explain its every secret, then have other craftsmen prove it could be copied, insisting that a single miraculous watch was useless if it could not be reproduced. It was a reasonable principle wielded, Harrison felt, as a weapon. He was made to take his life's work apart before their eyes. The old man, by now in his seventies, grew bitter, convinced the astronomers would never let a self-taught carpenter beat them at their own game. He was nearly right.

K1, Captain Cook, and Vindication

The proof that settled the argument was not Harrison's own watch but a copy of it. To satisfy the Board's demand that H4 be reproducible, a gifted London craftsman named Larcum Kendall was commissioned to build an exact duplicate. The result, known as K1, was itself a marvel, and it went to sea in the best possible hands. Captain James Cook carried K1 on his second great Pacific voyage, from 1772, and used it to fix his position day after day through some of the least-known waters on Earth. Cook grew to trust it completely, praising it in his journals as his never-failing guide, and even calling it, with sailor's affection, our trusty friend. When his astonishingly accurate charts were compared with the old reckoning, the case for the chronometer was closed.

Harrison's own vindication came grudgingly and late. Denied the full prize for years, he appealed over the Board's head, and King George the Third, who tested one of Harrison's watches personally at his private observatory and found it excellent, took up the old man's cause. By tradition the King declared that Harrison had been shamefully treated and would be put right, and in 1773 Parliament finally granted him a finishing sum that brought his lifetime reward close to the original twenty thousand pounds. He died three years later, in 1776, aged eighty-three, honoured at last.

The consequences outran the man entirely. Once chronometers could be built in numbers, and prices fell within a few decades, every serious ship carried one, set to Greenwich time, ticking steadily through storms and calms. Longitude ceased to be a deadly guess and became a glance at a dial. That is the deep reason the world's zero sits on this hill. Greenwich time, boxed in brass and carried into every ocean, became the reference against which the entire planet fixed its place. Harrison's obsession did not just save sailors. It quietly wired the map of the world to a clock.

Maritime Greenwich: Baroque Splendor and a Bullet Hole

The ensemble you're walking through — the Queen's House, the Old Royal Naval College, the park rising to the Observatory — is a UNESCO World Heritage Site, and its axis is a piece of royal deference: Wren designed the Naval College in two split halves specifically so it wouldn't block the Queen's House's view of the river. The Queen's House itself, begun by Inigo Jones in 1616, was England's first fully classical building — a piece of Italy dropped into Tudor Greenwich, so startling to contemporaries that it was called the House of Delight. Greenwich was a major royal palace before any of this: Henry the Eighth and Elizabeth the First were both born here. In the National Maritime Museum, the single most affecting object is Nelson's undress coat from Trafalgar, the twenty-first of October, 1805 — the fatal musket ball's entry hole visible at the left shoulder, the epaulette damaged, and by tradition it's displayed with the story that his decorations made him a mark for the French sharpshooter. The museum is free, so treat it as a raid rather than a campaign: the coat, the enormous Turner painting of Trafalgar — his largest work — and Yinka Shonibare's ship-in-a-bottle model of HMS Victory with sails of African-print fabric, which stood on Trafalgar Square's Fourth Plinth before coming home to Greenwich.

> Related listening: Christopher Wren designed both the cathedral and Greenwich's riverside masterpiece. Listen back — "St Paul's: Wren's Second City" (St Dunstan & St Paul's, Day 3). Related listening: Nelson's coat is at Greenwich; his column and lions preside over Trafalgar Square. Coming up — "St James's Park and Trafalgar Square: Power as Landscape" (Westminster Walk, Day 6).

The Painted Hall: Britain's Sistine Chapel

Step into the Painted Hall in the Old Royal Naval College and you enter what is only half-jokingly called the Sistine Chapel of Britain. Every surface, ceiling and walls alike, erupts in Baroque painting, and almost all of it is the work of one man, Sir James Thornhill, who laboured here for nineteen years, paid by the yard, so much for the ceiling, less for the walls. The programme is unabashed propaganda. At its heart sit King William and Queen Mary, trampling tyranny, presiding over Protestant liberty and British sea power, a painted argument that the political settlement of 1689 was blessed by heaven itself. Look for Thornhill's own self-portrait tucked at the edge, his hand reaching out, a painter quietly reminding his patrons he was owed.

Remember what this building actually was. It was Greenwich Hospital, founded by Royal charter as a home for aged and injured seamen of the Royal Navy, the naval twin of Chelsea's home for old soldiers, and Wren gave these pensioners a dining hall so magnificent that they were eventually barred from eating everyday meals in it. The splendour meant for broken sailors became a monument they were too grand to use.

The Hall's most solemn hour came in January 1806, when the body of Admiral Lord Nelson, killed at Trafalgar, lay in state here before enormous crowds who came to mourn the age's greatest hero. It is a short walk from the coat with the bullet hole in the museum to the room where the nation grieved the man who wore it.

Today the building leads a curious double life as Britain's most sought-after film location, backdrop to more than two hundred productions. The Painted Hall and the grounds have stood in for revolutionary France in Les Misérables, for the high seas in Pirates of the Caribbean, and for palaces across The Crown, Bridgerton and beyond. You may find the columns outside strangely familiar, having seen them a dozen times without ever knowing their name.

The Tulip Stairs, the Armada, and the Reckoning

Inside the Queen's House, seek out the Tulip Stairs, a spiral that winds upward with no central column, seeming to hang in the air. It was the first geometric self-supporting staircase in Britain, an Italian idea realised by Inigo Jones, and its wrought-iron scrollwork, more likely lilies than tulips, gives it its name. It is also the setting for one of Britain's most discussed ghost photographs. The story goes that in 1966 a visiting clergyman photographed the empty stairwell and, when the film was developed, found what appeared to be a shrouded figure or two climbing the steps. Sceptics blame a long exposure catching passing visitors; nobody has ever quite dispelled it. Judge for yourself as you climb.

Nearby hangs the Armada Portrait of Elizabeth the First, one of the most famous images in English history, the Queen serene amid storm-tossed ships, her hand resting on a globe, fingers over the Americas. It is less a likeness than a manifesto, a monarch laying claim to the oceans and the New World in a single confident gesture, painted at the dawn of England's sea empire.

And that is the thread worth holding as you leave Greenwich, because this place tells its maritime story with growing honesty. The National Maritime Museum increasingly asks harder questions of the age it celebrates, tracing the wealth of ships and ports back through the trade in enslaved people and the machinery of empire, refusing to let the romance of sail stand unexamined. It is a fuller, more grown-up version of the story than the one this collection once told.

Yet notice the deepest irony of the whole hill. Greenwich's most powerful export was never a cargo. It was standards. The second, the time zone, the map's grid of longitude, the shared idea of a single global now, all radiated from this quiet park to organise the entire planet. Empires ship goods and eventually lose them. Greenwich shipped agreement, and the world is still standing on its line.

Longitude and Latitude: Finding the Way at Sea

Standing beneath the copper hull of the Cutty Sark this morning, it is worth remembering that this ship was, above everything else, a machine for crossing empty ocean and arriving exactly where she meant to. A tea clipper that could not find her way was worth nothing. To understand how her captains threaded ten thousand miles of featureless water from Shanghai to London, you have to understand the two invisible lines that every navigator lives by, and why one of them was easy and the other very nearly impossible.

Latitude, the north-south measurement, came first and came easily, because nature hands it to you. Stand on deck at noon, measure how high the sun climbs above the horizon, correct for the day of the year, and you know how far you are from the equator. At night you can do the same with the Pole Star, which sits almost exactly above the North Pole and barely moves. The Greeks knew this. Ptolemy was drawing lines of latitude across his maps in the second century. A sailor in 1500, or in 1869, could fix his latitude within a few miles using a sextant and a table, and the tool for it, measuring the angle between a star and the horizon, is one of the loveliest instruments ever made.

Longitude, the east-west measurement, was a nightmare. There is no east-west equivalent of the Pole Star, nothing fixed in the sky to sight against, because the whole heavens wheel overhead as the Earth turns. And here is the crucial insight the whole problem turns on: longitude is really a measurement of time. The Earth spins through three hundred and sixty degrees in twenty-four hours, which means fifteen degrees of longitude every hour. So if you know what time it is where you are, by looking at your own local noon when the sun is highest, and you also know at that exact instant what time it is back at your home port, the difference between the two clocks tells you how far east or west you have sailed. One hour of difference is fifteen degrees. Four minutes is one degree. The mathematics is simple. The engineering was murderous.

The trouble was the second clock, the one that had to keep home-port time faithfully through months at sea. No clock on Earth could do it. Pendulum clocks, the most accurate timekeepers on land, were useless on a heaving deck, where the swing of the pendulum was ruined by every roll and pitch. Temperature swung from tropics to Cape Horn and changed the length of every metal part. Salt air corroded, damp swelled the oil. A clock that lost or gained even a few seconds a day would, over a two-month voyage, put a ship dozens or hundreds of miles from where it thought it was, and ships died of that error. In 1707 a British fleet, confident of its position and wrong about it, piled onto the rocks of the Isles of Scilly on a foggy October night and around fourteen hundred to two thousand sailors drowned. The nation was appalled.

So in 1714 Parliament passed the Longitude Act and offered a fortune, up to twenty thousand pounds, the equivalent of millions today, to anyone who could find a reliable way to determine longitude at sea. Two camps formed, and their rivalry is one of the great stories of science. On one side stood the astronomers, who believed the answer was in the sky. Their method, called lunar distance, used the Moon as a clock: the Moon moves against the background stars at a steady, predictable rate, so by measuring the angle between the Moon and a chosen star and consulting elaborate pre-computed tables, a navigator could work out the time at Greenwich and therefore his longitude. It worked, but it demanded a clear sky, a good view of the Moon, and around four hours of fearsome calculation for a single fix.

On the other side stood a Yorkshire carpenter and self-taught clockmaker named John Harrison, who thought the astronomers had it backwards. The answer was not to read the sky but to build a clock so perfect it could carry home-port time in a box. Over forty years he built a series of extraordinary machines, and if you climb the hill to the Royal Observatory this afternoon you can see them still ticking. His first three, H1, H2 and H3, were large and brilliant and not quite good enough. Then in 1759 he produced H4, which looks like an oversized pocket watch, and it changed everything. On a voyage to Jamaica it lost only about five seconds over eighty-one days, an accuracy no one had believed possible.

Harrison should have been hailed and paid at once. Instead the Board of Longitude, stacked with astronomers who favoured the lunar method, and in particular the Astronomer Royal Nevil Maskelyne, who was Harrison's rival and also one of his judges, moved the goalposts again and again. They demanded he hand over his designs, build copies, submit to more trials. Harrison, by now an old man, was nearly cheated of his reward, and only the personal intervention of King George the Third, who reportedly declared "By God, Harrison, I will see you righted," got him most of his money through a special act of Parliament in 1773. He died three years later. The clock had won, but only just.

The verdict of history came from Captain James Cook, who carried a copy of Harrison's watch, made by Larcum Kendall and called K1, on his second great Pacific voyage in the 1770s. Cook called it his "trusty friend" and "never-failing guide," and used it to chart coastlines with an accuracy that had never been seen. Within a few decades, as chronometers became cheaper and more numerous, every serious ship carried one or several. By the time Cutty Sark was launched in 1869 the marine chronometer was simply part of the furniture, kept below in gimbals to stay level, wound with ritual care, checked against time signals. Every day at sea her officers took a noon sight of the sun for latitude, compared their chronometer to work out longitude, and pricked the ship's position onto the chart. That daily ceremony, sextant and chronometer and a moment of arithmetic, was how she won her races.

And this brings us to why the world's clocks are set to this particular hill in south-east London. If longitude is the difference between your local time and the time at a reference meridian, then everyone has to agree on which meridian counts as zero. For a long time they did not. The French measured from Paris, others from their own capitals, and charts from different nations disagreed. The Royal Observatory at Greenwich had been founded in 1675 by King Charles the Second for exactly this purpose, "to find the so-much desired longitude of places for the perfecting the art of navigation," and generations of British astronomers had produced the star catalogues and almanacs that made the lunar method work. So many of the world's ships were already using charts based on Greenwich that in 1884, at a conference in Washington attended by twenty-five nations, Greenwich was voted the Prime Meridian of the world, longitude zero. The French abstained and sulked, keeping Paris time under a euphemism until 1911. But the decision held, and it is why the brass line you will straddle in the Observatory courtyard, one foot in the eastern hemisphere and one in the western, is the line from which the entire planet measures east and west.

There is a lovely modern footnote to stand on that line for. Satellite navigation, which now does instantly and invisibly what took Harrison forty years to make possible, does not quite agree with the old brass strip. The GPS zero of longitude sits about a hundred metres to the east, out across the courtyard and into the park, because the satellites reference a mathematical model of the whole Earth rather than the local vertical that the Victorian astronomers used with their telescopes. So there are, in a sense, two prime meridians a hundred metres apart, and you can walk from one age of navigation to the next in about ninety seconds.

That is the real cargo of this small hill. Cutty Sark carried tea and wool, but Greenwich exported something less tangible and far more valuable: agreement. The grid of latitude and longitude that lets any point on Earth be named by two numbers, the reference meridian that lets those numbers mean the same thing to everyone, the fusion of clock and star that turned a deadly guessing game into a routine, these are the quiet infrastructure of a connected world. When your phone tells you where you are tonight, it is finishing a sentence that a drowned fleet, a stubborn carpenter, and a racing clipper all helped to write.

The Story of Time: From Sundials to the Atomic Second

Everything you saw at Greenwich today was, in the end, about one question: what time is it, really? The red ball that drops from the Observatory turret at one o'clock, the brass meridian line, the room full of Harrison's clocks, the very phrase Greenwich Mean Time, all of it is the record of humanity's long struggle to measure time and, harder still, to make everyone agree on it. It is worth telling that story from the beginning, because it ends with the invisible machinery that runs your phone, your bank, and the satellites overhead tonight, and because so much of it was decided on this small hill.

For almost all of human history, time meant the sun. The first clock was a shadow. The Egyptians raised obelisks whose shadows swept across the ground through the day and built shadow clocks and sundials more than three thousand years ago. The logic was simple and universal: noon is the moment the sun stands highest and shadows are shortest, and the hours are slices of the sun's journey across the sky. The trouble with a sundial is obvious the moment a cloud passes or the sun sets, so people built clocks that did not need the sky. Water clocks, dripping steadily from one vessel to another, kept time in Egypt, Greece, China and the Islamic world for millennia; the Chinese engineer Su Song built a colossal water-driven astronomical clock tower in 1088. There were candle clocks marked in hours, and incense clocks that measured time by scent. But all of them drifted, and none could be trusted for long.

The great leap came in medieval Europe, and fittingly it came from the church. Monasteries needed to know when to ring the bells for prayer, and by the late thirteenth century mechanical clocks driven by falling weights, regulated by a clever oscillating mechanism called the verge-and-foliot escapement, were appearing in monastery and cathedral towers. These early clocks had no hands and no face; they simply struck a bell, and our word clock comes from the medieval word for bell. They were wildly inaccurate by our standards, losing or gaining perhaps fifteen minutes a day, but they were revolutionary, because for the first time time was being kept by a machine indoors, independent of the sun, ticking on through the night.

For four centuries clocks slowly improved, but the next enormous jump came in 1656, when the Dutch scientist Christiaan Huygens built the first pendulum clock. Galileo had noticed that a swinging weight keeps remarkably steady time, and Huygens turned the insight into an instrument. Overnight, accuracy improved from about fifteen minutes a day to around fifteen seconds a day, a hundredfold gain. The tall longcase clocks, the grandfather clocks, that followed were the most accurate machines on Earth, and for the first time it made sense to put a minute hand on a clock at all. But a pendulum needs to hang still and level, which is why, as this morning's story of longitude made clear, the pendulum was useless at sea, and why John Harrison's spring-driven marine chronometers of the 1760s were such a triumph. By carrying accurate time in a box across the ocean, the chronometer did not just save ships; it proved that portable, mechanical timekeeping could be trusted to a fraction of a second a day.

And here we arrive at the strange truth that Greenwich forces you to confront. For all these centuries, even as clocks grew superb, there was no such thing as the time. Every town kept its own local solar time, its clocks set so that noon fell when the sun was highest over that particular spot. Noon in Bristol was about ten minutes later than noon in London, because Bristol is further west and the sun reaches it later. Nobody minded, because nobody could travel fast enough for it to matter. A coach took days; the sun set your watch when you arrived. It was the railway that broke this ancient arrangement. When trains could carry you from London to Bristol in a couple of hours, a timetable written in dozens of different local times became chaos, and a real danger, because two trains running on clocks ten minutes apart can collide. So the railways imposed order. The Great Western Railway adopted London time along its whole line in 1840, and other companies followed, and this new synchronized standard was actually called railway time.

By the 1850s most of Britain had abandoned its local suns and set its clocks to a single national standard, and that standard was the mean time kept at the Royal Observatory here at Greenwich. It became law across Great Britain in 1880.

Why Greenwich? Because Greenwich had spent two hundred years becoming the most trusted timekeeper in the world. The Observatory, founded in 1675, existed to measure the heavens precisely enough to solve navigation, and precise astronomy demands precise time. So Greenwich distributed its time outward. From 1833 the great red time ball you saw has dropped down its mast at exactly one o'clock every afternoon, a visual signal so that ships' captains on the Thames could set their chronometers to the instant before sailing. On the Observatory wall hangs the Shepherd Gate Clock, installed in 1852, which showed Greenwich time to the public on a twenty-four-hour dial and was one of the first clocks ever driven by electric pulses sent down a wire. Time was becoming a signal, something you could send by telegraph across a whole country in the blink of an eye, and Greenwich was the source.

The final step was to do for the world what the railways had done for Britain. If Britain needed one time, a planet crossed by telegraph cables and steamships needed a shared system too. At the International Meridian Conference in Washington in 1884, the same conference that made Greenwich the Prime Meridian of longitude, the nations agreed to build the world's time zones outward from Greenwich. The globe was divided into twenty-four zones, each fifteen degrees of longitude wide, each one hour apart, all counted from the meridian that runs through the Observatory courtyard. This is why, wherever you are on Earth tonight, your time zone is defined as so many hours ahead of or behind Greenwich. Greenwich Mean Time became the still point around which the whole turning world set its clocks.

But the story does not end with mechanical clocks and Victorian confidence, because the twentieth century discovered that the Earth itself is an unreliable clock. In 1927 the quartz clock arrived, using the steady electrical vibration of a quartz crystal, and it was so stable it could actually detect tiny irregularities in the Earth's rotation, wobbles and a gradual slowing that no one had been able to measure before. The ground beneath the whole system, the assumption that one day is a fixed length, turned out to be slightly false. So physicists went looking for a clock that did not depend on the Earth at all, and they found it in the atom. In 1955, at the National Physical Laboratory in England, Louis Essen and Jack Parry built the first accurate atomic clock, which counts time by the fantastically regular vibration of caesium atoms, more than nine billion oscillations every second. It was so much better than the spinning Earth that in 1967 the world redefined the second itself: no longer a fraction of a day, but a fixed number of caesium vibrations, 9,192,631,770 of them. For the first time in history, time was cut loose from the sun and the stars.

This creates a delicate problem, and it is the reason the phrase Greenwich Mean Time is now, strictly speaking, slightly out of date. The atomic clocks keep perfect, unchanging time, but the Earth keeps wandering, spinning a hair slower over the centuries as the Moon's tides drag on it. So the world's official time, Coordinated Universal Time or UTC, is a compromise: it ticks with atomic precision, but every few years, when the gap between atomic time and the sun grows too wide, the world adds a leap second, holding the clocks back for one strange extra tick to let the sun catch up. Twenty-seven of these leap seconds have been quietly inserted since 1972. They are fiddly and they cause trouble for computers, and in 2022 the world's timekeepers voted to abolish them by 2035, finally letting atomic time and solar time drift gently apart. When people say GMT today, they almost always mean UTC, atomic time wearing Greenwich's old name.

And that atomic time is now the invisible heartbeat of modern life. The internet, the electricity grid, and the global financial system all depend on computers agreeing on the time to within thousandths of a second. Most of all, satellite navigation is simply timekeeping in disguise: each GPS satellite carries an atomic clock, and your phone finds its position by measuring, to within billionths of a second, how long signals take to arrive from several of them. It is so precise that engineers must correct for Einstein's relativity, because the clocks in orbit, moving fast and sitting higher in the Earth's weaker gravity, genuinely run about thirty-eight millionths of a second faster each day, and if that were ignored, GPS would drift you off course by miles within a day. The newest optical atomic clocks are so accurate they would not lose a single second over the entire age of the universe, and they may soon force us to redefine the second all over again.

So when you looked at the time ball dropping over Greenwich today, you were looking at the visible tip of an idea that runs from a shadow on Egyptian sand to a caesium atom humming in a laboratory. The sundial gave us the hours, the monks gave us the machine, Huygens gave us the pendulum, Harrison gave us portable time, the railways gave us a shared time, Greenwich gave the world its zero, and the atom gave us a second so perfect it no longer needs the sky at all. The whole of that history is why, in a small park in south-east London, you can stand on the line where the world agreed to start counting time.

> Related listening: From Greenwich's master clocks to Britain's most famous public clock. Coming up — "The Elizabeth Tower: Precision, Cracks and Pennies" (Westminster Walk, Day 6).